

Angela Radulescu

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Academic Positions

- 2022 – **Assistant Professor**, Icahn School of Medicine at Mt. Sinai (ISMMS)
Departments of Psychiatry and Neuroscience
Center for Computational Psychiatry
Lipschultz Center for Cognitive Neuroscience
Center for Integrative Systems and Neural Computation (InSyNC)
- 2020 – 2022 **Faculty Fellow/Assistant Professor**, New York University
Center for Data Science

Education

- 2014 – 2020 Ph.D. in Cognitive Psychology and Neuroscience
Princeton University
Advisers: Yael Niv, Nathaniel Daw
- 2007 – 2011 B.A. in Neuroscience and Behavior, Economics
Columbia University

Awards & Honors

- 2020 Schmidt Science Fellowship Finalist
- 2017 Travel Award, Reinforcement Learning and Decision Making (RLDM)
- 2017 Cognitive Science Graduate Fellowship, Princeton University
- 2017 Re-entry Fellowship, Prison Teaching Initiative at Princeton University
- 2015 Charlotte and Morris Tanenbaum *52 Fellowship, Princeton University
- 2014 Travel Award, Computational and Systems Neuroscience (Cosyne)
- 2013 Travel Award, Mechanisms of Motivation, Cognition and Aging Interactions
- 2011 Dean's List, Columbia University
- 2010 Summer Undergraduate Research Fellowship, Columbia University
- 2007 John Jay Scholar, Columbia University

Funding

- 2025 – BD² Discovery Research Grant - Intracranial Neurophenotyping of State Switches in Bipolar Disorder (Co-I)
- 2020 – 2023 Facebook Reality Labs, Cognitive Science Consortium Grant - Toward Rich User-World Predictive Interaction Models to Enable Human-Machine Collaboration (MPI)

Peer-Reviewed Publications [*: equal contribution, #: mentee]

- 2026 **Radulescu A***, van Opheusden B*, Callaway F, Griffiths TL, & Hillis, JM. (2026). A resource-rational account of human eye movements during immersive visual search. *Open Mind*, 10, 91-117.
- 2025 Trach JE#, deBettencourt MT, **Radulescu A**, McDougale SD. Rewards transiently and automatically enhance sustained attention. *Journal of Experimental Psychology: General*.
- 2024 Zhu J, **Radulescu A**, Bennett D. Emotional overshadowing: pleasant and unpleasant cues overshadow neutral cues in human associative learning. *Affective Science*, 5(3), 222-231.
- Li J#, **Radulescu A**. Dynamic self-efficacy as a computational mechanism of mania emergence. *Proceedings of the 46th Annual Conference of the Cognitive Science Society*.
- 2023 Bennett D*, **Radulescu A***, Zorowitz S, Falso V, Niv Y. Affect-congruent attention modulates generalized reward expectations. *PLOS Computational Biology*, 19(12), e1011707.
- Wise T, Emery K#, & **Radulescu A**. Naturalistic reinforcement learning. *Trends in Cognitive Sciences*, 28(2), 144-158.
- 2022 **Radulescu A**, Vong WK, Gureckis TM. Name that state: How language affects human reinforcement learning. *Proceedings of the 44th Annual Conference of the Cognitive Science Society*.
- 2021 **Radulescu A**, Shin Y, Niv Y. Human Representation Learning. *Annual Review of Neuroscience*, 44.
- 2020 Daniel R, **Radulescu A**, Niv Y. Multidimensional probabilistic learning reveals impaired attentional control during reinforcement learning in older adults. *Journal of Neuroscience*, 40(5), 1084-1096.
- 2019 **Radulescu A**, Niv Y. State representation in mental illness. *Current opinion in neurobiology*, 55: 160-166.
- Radulescu A**, Niv Y, Ballard IC. Holistic reinforcement learning: the role of structure and attention. *Trends in Cognitive Sciences*, 23(4), 278-292.
- 2017 Leong YC*, **Radulescu A***, Daniel R, DeWoskin V, Niv Y. Dynamic interaction between reinforcement learning and attention in multidimensional environments. *Neuron*. 93(2), 451-463.
- 2016 **Radulescu A**, Daniel R, Niv Y. The effects of aging on the interaction between reinforcement learning and attention. *Psychology and Aging*, 31(7), 747.
- Arkadir D, **Radulescu A**, Lubarr N, Raymond D, Bressman SB, Mazzoni P, Niv Y. *DYT1* dystonia increases risk taking in human. *eLife*, 5, e14155.
- 2015 Niv Y, Daniel R, Geana A, Gershman SJ, Leong Y, **Radulescu A**, Wilson RC. Reinforcement learning in multidimensional environments relies on attention mechanisms. *Journal of Neuroscience*, 35, 8145-8157.
- 2014 Gershman SJ, **Radulescu A**, Norman KA, Niv Y. Statistical computations underlying the dynamics of memory updating. *PLoS Computational Biology*, 10, e1003939.

Preprints [*: equal contribution, #: mentee]

- 2026 Maher C#, Qasim S, Tostaeva G, Nunez Martinez L, Ghatan S, Panov F, **Radulescu A*** & Saez I. (2026). Coordinated human prefrontal dynamics sustain task-state representations during learning. *bioRxiv*, 2026-05.
- Maher C#, Saez I, & **Radulescu A**. Decoding Covert Human Attention in Multidimensional Environments. *bioRxiv*, 2026-03.
- 2025 Beltran J#, ... & **Radulescu A**, Morris L. VTA network dominance in depression confers distinct psychopathological states through blunted neural tracking of reward prediction errors, 09 September 2025, available at *Research Square*.
- 2024 Chen Y, **Radulescu A**, Wu HZ. Unveiling the latent dynamics in social cognition with multi-agent inverse reinforcement learning. *bioRxiv*, 2024-10.
- 2020 **Radulescu A**, Holmes K, & Niv Y. (2020). On the convergent validity of risk sensitivity measures. *PsyArXiv*, 25(10.31234).

Peer-Reviewed Extended Abstracts [*: equal contribution, #: mentee]

- 2025 Maher C#, Saez I, **Radulescu A**. Decoding latent human attention across cognitive models. *Reinforcement Learning and Decision-Making (RLDM)*, Dublin, Ireland.
- Maher C#, Qasim S, Nunez L, Saez I*, **Radulescu A***. Intracranial recordings uncover neuronal dynamics of multidimensional reinforcement learning. *Computational and Systems Neuroscience (Cosyne)*, Montreal, Canada.
- Li J#, MK Ho*, **Radulescu A***. Act-Now vs. Look-Ahead: a dual β framework of self-efficacy in reinforcement learning. *Reinforcement Learning and Decision-Making (RLDM)*, Dublin, Ireland.
- 2024 Maher C#, Qasim S, Nunez Martinez L, Saez I*, **Radulescu A***. Intracranial recordings reveal neural encoding of attention-modulated reinforcement learning in humans. *Computational Cognitive Neuroscience (CCN)*, Boston, MA. [paper selected for a talk]
- Li J#, **Radulescu A**. A link between self-efficacy and optimistic overgeneralization. *Computational Cognitive Neuroscience (CCN)*, Boston, MA.
- Eckstein M, Miller KJ, **Radulescu A**. A Generative Grammar for Automatically Designing Experiments on Human Learning and Decision Making. *Computational Cognitive Neuroscience (CCN)*, Boston, MA.
- 2020 **Radulescu A***, van Opheusden B*, Callaway F, Griffiths TL, Hillis JM. From heuristic to optimal models in naturalistic visual search. *Bridging AI and Cognitive Science workshop (BAICS), International Conference for Learning Representations (ICLR)*, Addis Ababa, Ethiopia (online). [paper selected for a talk, 4/63 acceptance rate]
- 2019 **Radulescu A**, Niv Y, Daw ND. A particle filtering account of selective attention during learning. *Computational Cognitive Neuroscience (CCN)*, Berlin, Germany.

Davidson G[#], **Radulescu A**, Niv Y. Contrasting the effects of prospective attention and retrospective decay in representation learning. *Reinforcement Learning and Decision Making (RLDM)*, Montreal, Canada.

2018 Davidson G[#], **Radulescu A**, Niv Y. Passive forgetting or selective attention? Comparing two models of learning in multidimensional environments. *Computational Cognitive Neuroscience (CCN)*, Philadelphia, PA.

2017 **Radulescu A**, Leong YC, Niv Y. Reward sensitive attention dynamics during human reinforcement learning. *Computational Cognitive Neuroscience (CCN)*, New York, NY.

Radulescu A, Leong YC, Niv Y. Reward sensitive attention dynamics during human reinforcement learning. *Reinforcement Learning and Decision Making (RLDM)*, Ann Arbor, MI. [paper selected for a talk]

Radulescu A, Leong YC, Niv Y. Reward sensitive attention dynamics during human reinforcement learning. *Vision Sciences Society (VSS)*, St. Pete Beach, FL. [paper selected for a talk]

Hitchcock P, **Radulescu A**, Niv Y, Sims C. Building on solid ground: establishing the stability of computational modeling parameters. In Hitchcock, P. (Chair), *Introducing Computational Clinical Science: New Techniques to Improve Methods, Theory, Diagnosis, and Prediction*. Symposium to be presented at *51st Annual Convention for the Association for Behavioral and Cognitive Therapies*, San Diego, California.

Commentaries

2021 **Radulescu A**. Can data include personal narrative? *NYU Center for Data Science blog*. [\[link\]](#)

2014 Niv Y, Langdon AJ, **Radulescu A**. A free-choice premium in the basal ganglia. *Trends in Cognitive Sciences*, 19(1), 4-5.

Conference Poster Presentations (*selected*) [*: equal contribution, #: mentee]

2025 Beltrán JM[#], Mehta MM, Butler G, **Radulescu A**, Morris LS. Investigating the neural encoding of reward prediction errors and anhedonia using 7-Tesla MRI. *Society for Biological Psychiatry*, Toronto, Canada.

2025 Li J[#], **Radulescu A**. Dynamic self-efficacy as a computational mechanism for positive overgeneralization. *Society for Biological Psychiatry*, Toronto, Canada.

2024 Xie M[#], Gu X, **Radulescu A**. Effects of attention and learning on mood dynamics. *Computational Psychiatry Conference*, Minneapolis, MN.

2023 Beltrán JM[#], Mehta MM, Butler G, **Radulescu A**, Morris LS. Exploring the neural correlates of reward and punishment learning in depression using 7-Tesla MRI. *Computational Psychiatry Conference*, Dublin, Ireland.

Li J[#], **Radulescu A**. Dynamic self-efficacy updating as a computational mechanism of mania emergence. *Computational Psychiatry Conference*, Dublin, Ireland.

2022 Maher C[#], Gu X, **Radulescu A**, Saez I. The neural basis of representation learning in the human prefrontal cortex. *Society for Neuroscience Annual Meeting*, San Diego, CA.

- 2018 **Radulescu A**, Niv Y. Separable attention processes constrain multidimensional reinforcement learning. *Society for Neuroscience Annual Meeting*, San Diego, CA.
- 2017 Bu J#, **Radulescu A**, Turk-Browne NB, Niv Y. Feature-based reward learning biases dimensional attention. *Vision Sciences Society (VSS)*, St. Pete Beach, FL.
- Radulescu A**, Leong YC, Niv Y. Reward-sensitive attention dynamics during human reinforcement learning. *Computational Cognitive Neuroscience*, New York, NY.
- 2016 **Radulescu A**, Allefeld C, Schuck N, Haynes JD, Niv Y. Studying value-guided decision making through model-based multivariate fMRI analysis. *Society for Neuroeconomics*, Berlin, Germany.
- 2015 **Radulescu A**, Niv Y. Learning state representations from experience. (2015). *Machine Learning Summer School*, Tübingen, Germany.
- 2014 Arkadir D, **Radulescu A**, Lubarr N, Raymond D, Bressman SB, Mazzoni P, Niv Y. A link between corticostriatal plasticity and risk taking in humans. *Computational and Systems Neuroscience (Cosyne)*, Salt Lake City, UT. [presenting author]
- 2012 **Radulescu A**, Niv Y. Age-related differences in learning to selectively attend. *Society for Neuroscience Annual Meeting*, New Orleans, LA.

Invited Talks

- 2026 **Mathematics of Neuroscience and AI**
Current Directions in Neuroscience & the Law, Columbia University
- 2025 **Reinforcement Learning and Decision-Making (RLDM) Conference**
- 2024 **Columbia University Seminar on Cognitive and Behavioral Neuroscience**
SUNY Downstate Behavioral and Neural Science Seminar
- 2023 **Max Planck UCL Center for Computational Psychiatry**
SfN Minisymposium: Generalization for Learning and Decision-Making
NIA Workshop on Computational Approaches to Advance Aging and AD/ADRD Research
Algorithms for Building and Structuring Internal Models, Park City Winter Conference
- 2022 **NSF/Simons NeuroDataScience Workshop, University of California, Irvine**
Maps in Reinforcement Learning RLDM Conference Workshop
NYU Center for Data Science, Women in Data Science Panel
Dartmouth College Cognitive Brown Bag
University of Chicago Department of Psychology Cognition Workshop
- 2021 **RLDM Meeting, Max Planck Institute, Tübingen, Germany**
University of California, Berkeley Department of Psychology
Department of Psychiatry, Icahn School of Medicine at Mt. Sinai
Shenhav Lab, Cognitive, Linguistic & Psychological Sciences, Brown University

- 2020 **Max Planck UCL Center for Computational Psychiatry**
Microsoft Research Seminar
MILA, Neural-AI Reading Group
- 2019 **New York University, Department of Psychology, ConCats Seminar**
Princeton Neuroscience Institute Retreat
Rutgers University, Cognitive Science Graduate Seminar
Tel Aviv University, Department of Psychology
- 2018 **Manhattan Area Memory Meeting**

Invited Workshops and Tutorials

- 2025 **School on Analytical Connectionism.** Bias in human reinforcement learning lecture
- 2024 **SRDNA Computational Modeling Workshop.** Reinforcement learning tutorial
- 2024 **Computational Psychiatry Conference.** Reinforcement learning tutorial
- 2021 **FLUX Computational Modelling Workshop.** Computational models of human gaze data tutorial
- 2021 **Mental Effort Workshop.** Computational models of human gaze data tutorial

Teaching

- 2024 – **New York Computational Psychiatry Workshop** – founding co-director
- 2023 – **Modern Statistics for Modern Biology**, ISMMS – co-director
- 2022 – **Fundamentals of Computational Psychiatry**, ISMMS – founding co-director
- 2022 – **Selected Topics in Neuroscience**, ISMMS – co-director
- 2021 **Advanced Topics in Data Science**, New York University – co-lead instructor
- 2018 **Introduction to Cognitive Neuroscience**, Princeton University – assistant instructor
- 2015 – 2019 **Introduction to Psychology**, Princeton Prison Teaching Initiative – instructor, team lead
- 2017 **Introduction to Neuroscience**, Columbia University High School Programs – instructor
- 2015 **Developmental Psychology**, Princeton University – assistant instructor
- 2010 **Thinking and Decision Making**, Columbia University – teaching assistant

Mentoring

- 2026 – Meghan Gallo (postdoctoral fellow at Columbia)
- 2026 Luca Kosina (MA student, Tübingen University)
- 2023 – 2025 Marjorie Xie (NYAS AI and society fellow, ISMMS → postdoc at NYU)
- 2023 – Jing Li (graduate student, neuroscience, ISMMS)
- 2022 – 2025 Catherine Kim (research coordinator, ISMMS → Matter Neuroscience)
- 2022 – 2026 Jackie Beltran (graduate student, neuroscience, ISMMS → Havas Health Medical Fellowship)
- 2022 – 2023 Itzel Martinez (MA student, neuroscience, ISMMS)
- 2021 – 2026 Christina Maher (graduate student, neuroscience, ISMMS → Meta Reality Labs)
- 2021 – 2023 Kara Emery (postdoctoral fellow, NYU Center for Data Science → NYU McSilver Institute)
- 2021 Sally Leung (NYU undergraduate independent research → MA student at Columbia)
- 2021 Praxal Patel (NYU Center for Data Science MA independent research)
- 2018 Guy Davidson (Princeton summer internship → graduate student at NYU Center for Data Science)
- 2017 – 2018 Julie Newman (Princeton undergraduate senior thesis → ClearView Healthcare)

2015 – 2017 Jennifer Bu (Princeton undergraduate senior thesis → medical student at UCSD)
2015 David Wang (Princeton summer internship → medical student at Stanford)

Thesis committees

2026– Natalie McClain (ISMMS)
2025– Ross Kempner (ISMMS)
2023–2026 Alessandra Yu (ISMMS)
2023 Matthew Schafer (ISMMS)
2023–2026 Qixiu Fu (ISMMS)
2023 Nathan Tyler-Hall (UNC Chapel Hill)
2023–2026 Alexandra Fink (ISMMS)
2023–2025 Pushkala Jayaraman (ISMMS)

Ad-hoc Reviewer

ACM Symposium on Eye-tracking Research and Applications (ETRA) | *Aging, Neuropsychology and Cognition* | *Biological Psychiatry* | *Cerebral Cortex* | *Cognition* | *Cognitive, Affective and Behavioral Neuroscience (CABN)* | *Collabra* | *Communications Biology* | *Computational Cognitive Neuroscience (CCN)* | *Computational Systems Neuroscience (Cosyne)* | *Computational Psychiatry Conference (CPC)* | *Cognitive Science Society (CogSci)* | *Current Biology* | *Current Opinion in Behavioral Sciences* | *eLife* | *eNeuro* | *Journal of Cognitive Neuroscience* | *Journal of Mathematical Psychology* | *Journal of Neuroscience* | *Nature Communications* | *Nature Human Behavior* | *Neural Networks* | *Neurons, Behavior, Data Analysis, and Theory* | *NSF Division of Information and Intelligent Systems* | *PLoS Computational Biology* | *PNAS* | *Psychonomic Bulletin & Review* | *Science* | *Scientific Reports*

Service

2024 – **Icahn School of Medicine at Mt. Sinai (ISMMS)**, Curriculum Committee
2023 – **Computational Cognitive Neuroscience (CCN)**, Program Committee
2024 – 2025 **Computational Cognitive Neuroscience (CCN)**, DEI chair

Summer Courses

2025 **Neuro4Pros**, Queen's University, Kingston, ON, Canada
2018 **Methods in Neuroscience at Dartmouth (MIND)**, Dartmouth University, Hanover, NH
2015 **Machine Learning Summer School (MLSS)**, MPI Tübingen, Germany

Research Positions

2019 – 2020 Ph.D. intern, Facebook Reality Labs; manager: James Hillis
2011 – 2014 Lab manager, Princeton Neuroscience Institute; adviser: Yael Niv
2010 – 2011 Undergraduate research assistant, Columbia University; adviser: Jacqueline Gottlieb
2009 Undergraduate research assistant, Columbia University; adviser: Elke Weber